MODULE 22 · SECTION 3 - TECHNIQUES TO IMPROVE GEN AI MODEL OUTPUT

Sampling parameters and safety settings

The dials that shape a model's output: tokens, temperature, top-p, top-k, output length and safety filters, and which dial fixes which symptom.

SECTION 3 of 4 EXAM WEIGHT ~20% MODULE 22 of 33

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Same model, different dials. **Match the dial to the symptom**, and know which symptoms no dial can fix.

The controls, at a glance

CONTROL	WHAT IT SHAPES
Token count	How much goes in and out: cost, speed, fit
Temperature	How adventurous each word choice is
Top-p and top-k	How many candidate words are considered
Max output tokens	How long the reply may be
Safety settings	Which harmful content is blocked

ON THE EXAM

Every question in this area names a symptom. Find the dial that controls that symptom, and only that one.

Tokens: the unit of everything

- **A token** is a piece of a word; roughly three-quarters of a word in English.
- **Input tokens** include the prompt, instructions, documents and chat history.
- **Output tokens** are the reply, plus any thinking the model does.
- **Tokens drive** cost, response time, and whether it fits the context window.
- **Count tokens** before sending, to predict cost and avoid overflow.

THE SIGNAL

Cost or speed problems at scale usually trace back to token count. Trim the input, or cap the output.

Temperature: how adventurous each word is

- **The model scores** every possible next word by likelihood.
- **Low temperature:** it nearly always picks the most likely word.
- **High temperature:** less likely words get a real chance.
- **Low** means consistent and focused; **high** means varied and creative.
- **Temperature changes style,** not knowledge. It cannot add facts.

> THE TIEBREAKER
>
> Answers vary too much: lower temperature. Answers too bland or repetitive: raise it.

Low or high temperature?

LOWER

Consistent and focused

- Policy and factual answers
- Extracting fields into a fixed format
- Classification and tagging
- Same question, same answer

HIGHER

Varied and creative

- Campaign ideas and slogans
- Brainstorming names
- Several distinct drafts
- Fresh wording each time

THE TIEBREAKER

Facts, rules and structured output lean low. Ideas and creative drafts lean high.

Top-k and top-p: how many words are in the running

- **Top-k:** consider only the k most likely next words.
- **Top-k of 1** always takes the single most likely word.
- **Top-p:** consider the smallest set of words whose likelihoods add up to p.
- **Top-p of 0.9** keeps the words covering 90% of the probability.
- **Lower values** mean safer, narrower choices; higher values mean more variety.

REMEMBER

Top-k counts words. Top-p adds up probability. Both narrow the shortlist before temperature picks from it.

Max output tokens and stop sequences

- **Max output tokens** caps the length of the reply.
- **Too low** and answers stop mid-sentence.
- **Too high** allows rambling, cost and delay.
- **Thinking models** count their reasoning too; a low cap can cut it short.
- **Stop sequences** end the reply at a chosen marker.

THE SIGNAL

Answers cut off mid-sentence: raise max output tokens. Answers too long or costly: lower it, and ask for brevity.

Safety filters: categories and thresholds

SETTING	WHAT IT MEANS
Categories	Harassment, hate speech, sexually explicit, dangerous
Block most	Blocks low, medium and high probability of harm
Block some	Blocks medium and high probability
Block few	Blocks only high probability
Block none or off	No blocking in that category

REMEMBER

Filters score the probability of harm, not its severity. Child-safety protections are always on and cannot be changed.

What has changed with newer models

- **Gemini 3 guidance:** keep temperature at its default of 1.0.
- **Google warns** that lowering it can cause looping or weaker reasoning.
- **Safety defaults:** newer Gemini models default to no blocking in these categories.
- **So set thresholds deliberately**, and test them, for customer-facing apps.
- **The exam still tests** the classic rule: lower temperature, more consistency.

> CURRENCY CHECK
>
> Answer exam questions with the classic rules. In real projects, read the current model guidance and test.

Which dial fixes which symptom

SYMPTOM	DIAL
Same question, different answers	Lower temperature or top-p
Bland, repetitive ideas	Raise temperature
Replies cut off mid-sentence	Raise max output tokens
Too long, too costly	Lower max output tokens
Harmful content gets through	Stricter safety threshold
Wrong or invented facts	No dial: ground it

THE TIEBREAKER

The last row is the classic trap. A dial changes how the model writes, never what it knows.

Wrenfield's settings, by job

JOB	SETTINGS
Returns policy answers	Low temperature; grounded; moderate length
Autumn campaign slogans	Higher temperature; several drafts
Tagging products into JSON	Low temperature; fixed format; stop sequence
Customer chat	Safety thresholds set and tested; length capped

THE TAKEAWAY

Different jobs on the same model need different settings. One configuration for everything is a warning sign.

> Wrenfield's policy assistant is grounded in the correct handbook.
>
> But asked the same question twice, it phrases the answer very differently, and managers find that confusing.

Which setting change helps most?

A Raise the temperature

B Raise max output tokens

C Lower the temperature

D Turn off the safety filters

◆ PAUSE AND DECIDE

Which setting change helps most?

A Raise the temperature

More variety, the opposite of the goal.

B Raise max output tokens

Longer replies, not steadier ones.

C Lower the temperature

More consistent wording for the same question.

D Turn off the safety filters

Unrelated, and risky.

THE SIGNAL

Same question, different answers: lower temperature.

> Wrenfield's marketing team asks for 20 slogan ideas for the autumn campaign.
>
> The ideas come back safe, similar and dull.

Which setting change helps most?

A Raise the temperature

B Set top-k to one

C Lower the temperature

D Choose a stricter safety threshold

◆ PAUSE AND DECIDE

Which setting change helps most?

A Raise the temperature

Gives less likely, fresher words a chance.

B Set top-k to one

Always the single most likely word: even blander.

C Lower the temperature

Even more predictable.

D Choose a stricter safety threshold

Blocks content; it does not add creativity.

THE SIGNAL

Bland, repetitive ideas: raise temperature.

> Wrenfield's assistant writes delivery instructions for two-person sofa deliveries.
>
> Many replies stop mid-sentence, partway through the steps.

Which setting should change?

A Lower the temperature

B Choose a stricter safety threshold

C Lower top-k

D Raise max output tokens

◆ PAUSE AND DECIDE

Which setting should change?

A Lower the temperature

Changes wording, not length.

B Choose a stricter safety threshold

Would block more, not finish replies.

C Lower top-k

Narrows word choice; length is unchanged.

D Raise max output tokens

Gives the reply room to finish.

THE SIGNAL

Replies cut off mid-sentence: raise max output tokens.

> Wrenfield's product assistant refuses to describe a set of kitchen knives, flagging them as dangerous content.
>
> The team must sell knives, but still wants protection in every other category.

What is the best response?

A Turn off all safety filters

B Relax only the dangerous-content threshold, then test

C Raise the temperature

D Switch to a larger model

◆ PAUSE AND DECIDE

What is the best response?

A Turn off all safety filters
Removes protection everywhere.

B Relax only the dangerous-content threshold, then test
A targeted change, checked before launch.

C Raise the temperature
Changes wording, not filtering.

D Switch to a larger model
Filters are settings, not model size.

THE SIGNAL

Legitimate content blocked in one category: adjust that category, then test.

> Wrenfield's returns assistant already runs at a low temperature.
>
> It still sometimes states return rules that are not in any Wrenfield policy.

What should the team do?

A Ground answers in the current policy documents

B Lower the temperature to zero

C Lower top-p

D Raise max output tokens

◆ PAUSE AND DECIDE

What should the team do?

A Ground answers in the current policy documents

Gives the model the real rules to answer from.

B Lower the temperature to zero

Steadier wording, still invented rules.

C Lower top-p

Narrower words, still invented rules.

D Raise max output tokens

Longer answers, same invention.

THE TAKEAWAY

Wrong or invented facts: no dial fixes it. Ground it.

> An exam question describes a setting that limits the model to the smallest group of likely next words whose probabilities add up to a set threshold, such as 0.9.

Which setting is described?

A Temperature

B Top-k

C Max output tokens

D Top-p

◆ PAUSE AND DECIDE

Which setting is described?

A Temperature

Reshapes likelihoods; no threshold of total probability.

B Top-k

Counts a fixed number of words, not a total probability.

C Max output tokens

Caps length, not word choice.

D Top-p

The smallest set whose probabilities reach p.

THE SIGNAL

Adds up probability to a threshold: top-p. Counts words: top-k.

Module 22 in five lines

- Tokens drive cost, speed and fit; count them before you send.
- Temperature: low for consistency, high for creativity. It never adds facts.
- Top-k counts candidate words; top-p adds up their probability.
- Max output tokens caps length; safety thresholds block harm by category.
- Wrong facts are not a dial problem: ground the model.